

Bevel Gear Design

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1 Introduction

1.1 Bevel Gears Overview

A bevel gear is a gear that has a conical shape and is designed to transmit power and motion between two shafts which intersect at usually a 90-degree angle. The teeth in both gears allow them to change direction of rotation and torque. They are mainly used in automobiles, specifically in differentials to let wheels rotate at different speeds during turns.

Bevel gearboxes come with a couple of essential components. They mainly consist of the following:

1. The **bevel gear** itself. This transmits power and the required torque.
2. The **input shaft**. This receives power or torque from the input source and transmits it to the gears.
3. The **output shaft**. This delivers the modified power to the next part of the system
4. **Bearings**. They offer support and balance to the shafts reducing friction and offer smooth alignment and rotation.
5. The **lubrication**. This reduces wear in bearings and shafts
6. The **housing**. This protects the gears and everything from the outside environment.
7. **Seals and gaskets**. They prevent oil leaks and also help protect from outside environment

1.2 Project Overview

This project is mainly focused on designing a bevel gear box. All the components, calculations and other factors will have to be taken into consideration.

The main objectives, given the chosen gear sets, will be to calculate the forces, stresses, sealing and overall design of the gearbox including the housing as well. The selection of every material and arrangement is important and will be documented in this document.

2 Requirements

After selecting the gear ratio according to article number: 361 144 800. The table below summarises the data given in the article. The figure and the table below show the specifications according to the article number.

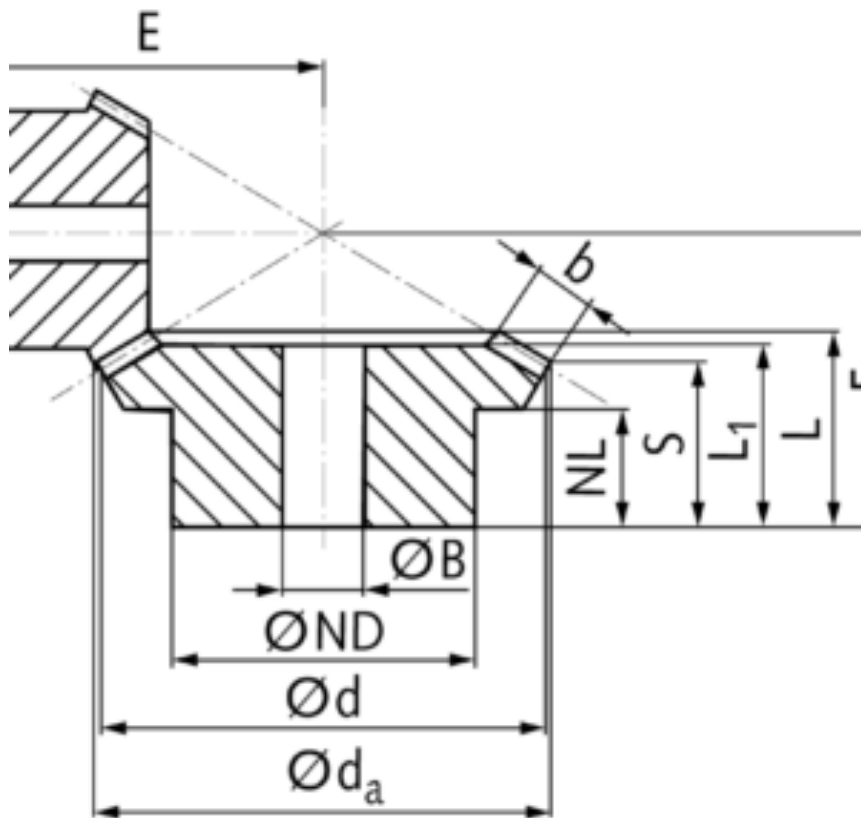


Figure 1: Gear specifications

Transmission [i]	1,5:1 (16/24)
Module	3
No. of Teeth	16
d_a [mm]	53,2
d [mm]	48
ND [mm]	40
NL [mm]	13,2
L_1 [mm]	30
L [mm]	32,7
S [mm]	17,7
b [mm]	19
B^{H7} [mm]	15
E [mm]	52
Admissible MD [Nm]	5,9
Weight [g]	250
Corresponding Counter Wheel [Article-No.]	36144900

Figure 2: Configurations for 36144800 (driving gear)

Transmission [i]	1,5:1 (16/24)
Module	3
No. of Teeth	24
d_a [mm]	75,5
d [mm]	72
ND [mm]	50
NL [mm]	8
L_1 [mm]	24
L [mm]	27,8
S [mm]	18,6
b [mm]	19
B^{H7} [mm]	15
E [mm]	40
Admissible MD [Nm]	8,9
Weight [g]	490
Corresponding Counter Wheel [Article-No.]	36144800

Figure 3: Configurations for 36144900 (driven gear)

3 Concept Design and Calculations

3.1 Calculations

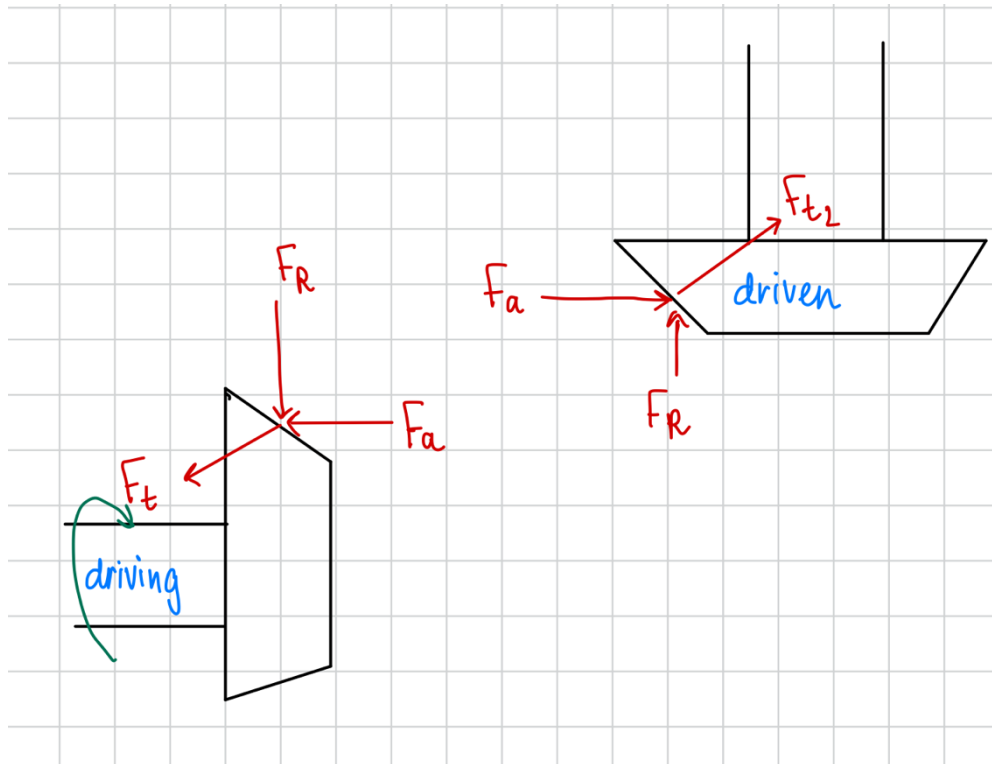


Figure 4: Free Body Diagram of Contact Forces

Firstly, to determine the pitch angle between the driver and driven shafts.

Using:

$$\text{Pitch cone angle } (\delta) = \frac{\text{No. of Teeth in Pinion}}{\text{No. of Teeth in Gear}}$$

$$\delta_1 = \frac{16}{24} \approx 0.6667 = \tan^{-1}(0.6667) \approx 33.69^\circ \text{ (Driver Gear)}$$

$$\delta_2 = 90^\circ - 33.69^\circ = 56.31^\circ \text{ (Driven Gear)}$$

After drawing the free body diagram introducing the contact forces, we can calculate the forces. Starting with tangential force, F_{t1} . Given the Pitch Diameter is 40mm.

Driving Pinion

Using the equation:

$$F_{t1} = \frac{2T}{P_D}$$

Where T is the torque, which gives us;

$$F_{t1} = \frac{2 * T}{P_D} = \frac{2 * 5.9Nm}{0.048m} = \mathbf{245.83\ N}$$

Assuming the normal pressure angle is 20° (**Maedler**), It can be used to calculate the radial force F_r ,

$$\begin{aligned} F_r &= F_{t1} * \tan(\theta) * \cos(\delta) \\ &= 245.83N * \tan(20) * \cos(33.69) \\ &\approx \mathbf{74.44\ N} \end{aligned}$$

Then we calculate the axial force F_a

$$\begin{aligned} F_a &= F_{t1} * \tan(\theta) * \sin(\delta) \\ &= 245.83N * \tan(20) * \sin(33.69) \\ &\approx \mathbf{49.63\ N} \end{aligned}$$

3.2 Shaft Stresses

According to the design, the **driving shaft** is made of **C45 steel** with a **15mm diameter** and a total length of **100mm**. The **driven shaft** is made of **42CrMo4 steel**, primarily **20mm in diameter and 150mm long**, but it includes an enlarged **24mm diameter section** that is **56.99mm long** to accommodate for bearing placement. Both the shafts should be strong enough to handle the forces calculated above. To make ensure that, the permitted stresses have to be calculated.

Driving Shaft

Bending Moment (M_{b1})

$$M_{b1} = F_{t1} * \frac{L}{2} = 245.83N * \frac{100mm}{2} = \mathbf{12\ 291.66\ Nmm}$$

Assuming the application factor (K_a) = 1.5,

Bending Stress (σ_{b1})

$$\sigma_{b1} = \frac{32 * M_{b1} * K_a}{\pi * d^3} = \frac{32 * 12291.66Nmm * 1.5}{\pi * (15mm)^3} = \mathbf{55.64\ MPa}$$

For C45 Steel, the allowable yield strength is **245MPa**, thus our stress is below that, thus it is within allowable stress limit.

Also, assuming the application factor (K_a) = 1.5,

Torsional Stress (τ_{s1})

$$\tau_{s1} = \frac{16 * T}{\pi * d^3} = \frac{16 * 5900Nmm * 1.5}{\pi * (15mm)^3} = \mathbf{13.35\ MPa}$$

Again, since the allowable yield strength is 245MPa, and the stress is way below that, it is safe. Now, to the von Mises equivalent stress,

$$\sigma_{eq,1} = \sqrt{\sigma_{b1}^2 + 3\tau_{s1}^2} = \mathbf{60.25\ MPa}$$

Which again is good since it way below the allowable yield strength.

Driven Shaft

$$F_{t2} = \frac{2T}{P_D} = \frac{2 * 8.9Nm}{0.072m} = \mathbf{247.22 \text{ N}}$$

Bending Moment (M_{b2})

$$M_{b2} = F_{t2} * \frac{L}{2} = 247.22N * \frac{150mm}{2} = \mathbf{18541.50 \text{ Nmm}}$$

Assuming the application factor (K_a) = 1.5,

Bending Stress (σ_{b2})

$$\sigma_{b2} = \frac{32 * M_{b2} * K_a}{\pi * d^3} = \frac{32 * 18541.50Nmm * 1.5}{\pi * (20mm)^3} = \mathbf{35.41 \text{ MPa}}$$

For 42CrMo4 Steel, the allowable yield strength is 500MPa, thus our stress is below that thus it is within allowable stress limit.

Also, assuming the application factor (K_a) = 1.5,

Torsional Stress (τ_{s2})

$$\tau_{s2} = \frac{16 * T}{\pi * d^3} = \frac{16 * 8900Nmm * 1.5}{\pi * (20mm)^3} = \mathbf{8.49 \text{ MPa}}$$

Again, since the allowable yield strength 500MPa, and the stress is way below that, it is safe. Now, to the von Mises equivalent stress,

$$\sigma_{eq,2} = \sqrt{\sigma_{b2}^2 + 3\tau_{s2}^2} = \mathbf{38.34 \text{ MPa}}$$

Which again is good since it way below the allowable yield strength.

3.3 Bearing Calculation

To select the right bearing, we must first calculate the forces to ensure the selected bearing can handles the radial and axial forces.

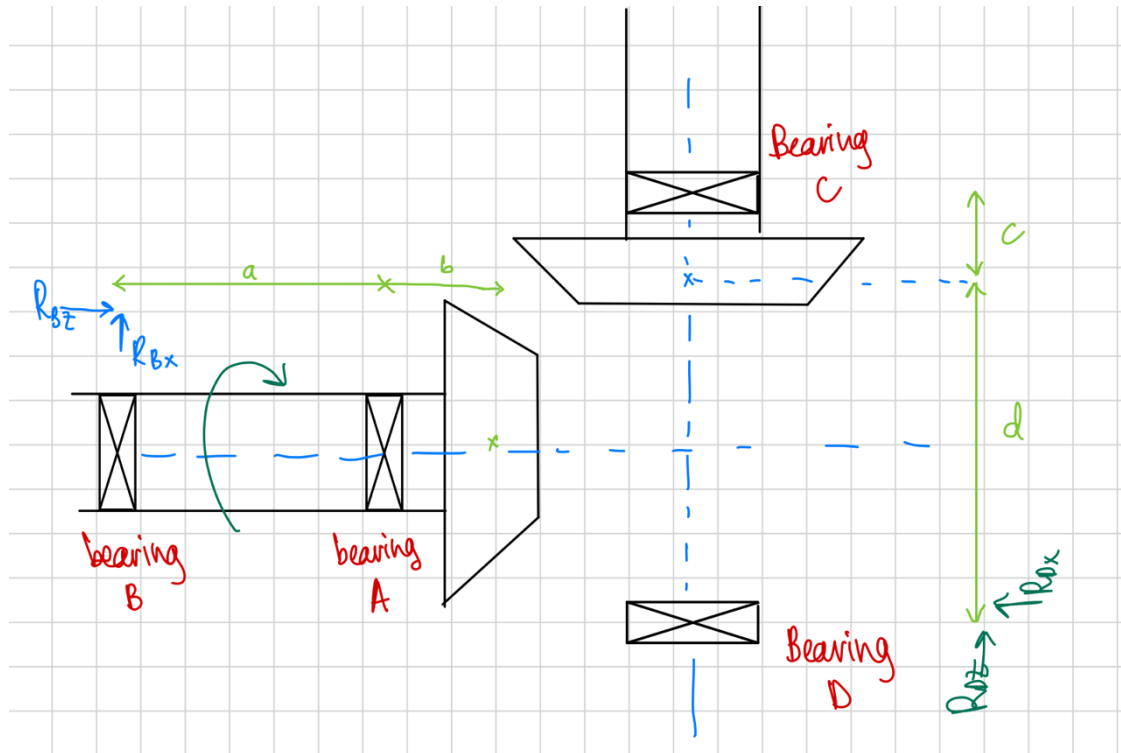


Figure 5: Free Body Diagram of Bearing Forces

$$a=42\text{mm}$$

$$b=23\text{mm}$$

$$c=18.3\text{mm}$$

$$d=78.2\text{mm}$$

$$\text{bearing b thickness} = \text{bearing a thickness} = 13\text{mm}$$

$$\text{bearing c thickness} = \text{bearing d thickness} = 18\text{mm}$$

Bearing B (floating)

Radial Force:

$$R_{bx} = \frac{F_r \cdot (a + b)}{a} = \frac{74.4 \cdot 65}{42} = \mathbf{115.14\ N}$$

Axial Force:

$$R_{bz} = 0$$

Resultant Force:

$$R_b = \mathbf{115.14\ N}$$

Bearing A (fixed)

Radial Force:

$$R_{ax} = F_r - R_{bx} = 74.4 - 115.14 = \mathbf{-40.74\ N\ (compression)}$$

Axial Force:

$$R_{az} = \mathbf{49.63\ N\ (compression)}$$

Resultant Force:

$$R_a = \sqrt{R_{Ax}^2 + R_{Az}^2} = \mathbf{64.20\ N}$$

Bearing D (floating)

Radial Force:

$$R_{dx} = \frac{F_r \cdot (d + c)}{d} = \frac{74.4 \cdot 96.5}{78.2} = \mathbf{91.81\ N}$$

Axial Force:

$$R_{dz} = 0$$

Resultant Force:

$$R_d = \mathbf{91.81\ N}$$

Bearing C (fixed)

Radial Force:

$$R_{cx} = F_r - R_{dx} = 74.4 - 91.81 = \mathbf{-17.41\ N\ (compression)}$$

Axial Force:

$$R_{cz} = \mathbf{49.63\ N\ (compression)}$$

Resultant Force:

$$R_c = \sqrt{R_{cx}^2 + R_{cz}^2} = \mathbf{52.59\ N}$$

3.4 Bending Moment Diagram

Since we know the forces, we can now do the bending moment diagram. The following bending moment diagram is for the driving shaft.

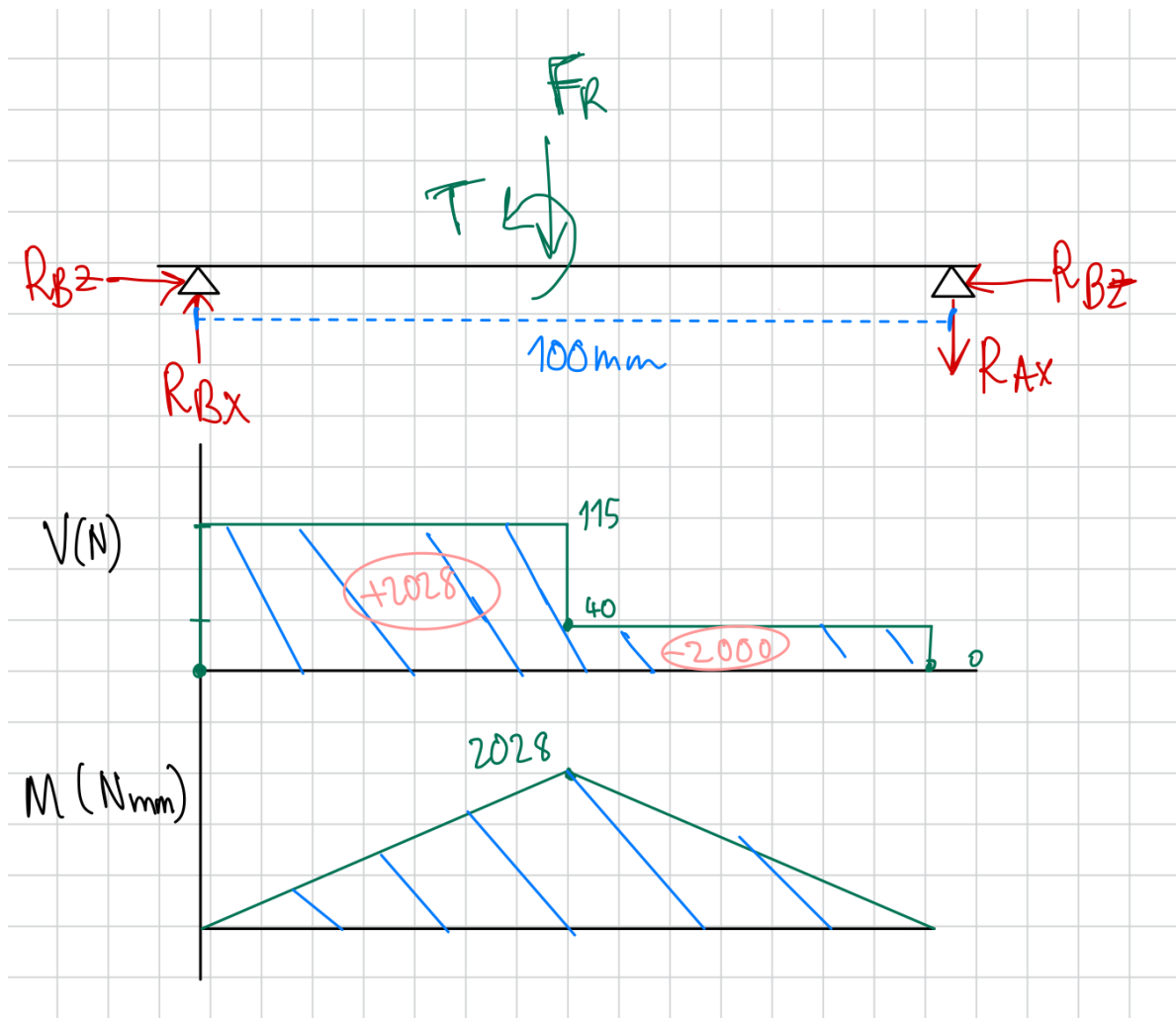


Figure 6: Bending Moment Diagram

3.5 Bearing selection

After calculating the forces acting on each bearing, the appropriate bearing types can be selected. The chosen bearings must not only support the loads generated by the bevel gear transmission but also provide sufficient service life under the expected operating conditions.

Bearings **A** and **C** are positioned closest to the pinion and the gear, respectively. These locations are subject to both **radial and axial loads**, and therefore require bearings capable of supporting combined loading.

For this reason, **tapered roller bearings** were selected for these positions. Specifically, **bearing 32204** was chosen for position A and **bearing 30302** for position C. These bearings are well-suited for handling simultaneous axial and radial forces. Detailed technical specifications for these bearings are provided in the following figures.

In contrast, **Bearings B and D** are located further away from the gear mesh and are designed to act as **floating supports**. As such, they primarily experience **radial loads** and must be free to accommodate axial displacement caused by thermal expansion or assembly tolerances.

For these positions, **deep groove ball bearings** were selected due to their reliability in supporting radial loads and their simple design. These bearings ensure smooth operation without constraining axial movement.

The bearings must support our axial and radial forces:

$$F_r \approx 74.44 \text{ N}$$

$$F_a \approx 49.63 \text{ N}$$

Driven Shaft Bearing (Bearing C)

Technical Data

Inside Ø (mm):	20
Outside Ø (mm):	47
Width (mm):	19.25
Width Inner Ring (mm):	18
Width Outer Ring (mm):	15
max. Operating Temperature:	+150°C
min. Operating Temperature:	-40°C
Tolerance for inside Ø (mm):	0/-0,012
Tolerance for outside Ø (mm):	0/-0,014
Tolerance For Width Outer Ring (mm):	0/-0,12
Tolerance for width of the bearing (mm):	0,2/0
Tolerance For Width Inner Ring (mm):	0/-0,12
Seal:	open
Ring Material:	Rolling Bearing Steel
Rolling Element Material:	Rolling Bearing Steel
Cage Material:	Rolling bearing steel
Type of lubrication:	oiled
Magnetic:	yes
Norm:	DIN 720
Item weight:	0,16 kg

Figure 7: Bearing 32204

Equivalent Load:

For bearing C, the dynamic loading is **27000N**, from the dataset, these are also the load factors given,

$$X = 0.56$$

$$Y = 1.55$$

But to account for real life scenarios, we can use other factors:

$$\text{Application Load } f_0 = 1.1$$

$$\text{Dynamic Radial } f_{rd} = 1$$

$$\text{Operating Temp } f_{tr} = 1$$

$$\text{Dynamic Axial } f_{ad} = 1.2$$

Then we can use these to calculate the dynamic load for **bearing 32204**

$$F_{r \text{ dynamic}} = f_0 * f_{rd} * f_{tr} * f_{ad} * F_r = 1.1 * 1 * 1 * 1.2 * 17.41 \text{ N} = \mathbf{22.98 \text{ N}}$$

$$F_{a \text{ dynamic}} = f_0 * f_{rd} * f_{tr} * f_{ad} * F_a = 1.1 * 1 * 1 * 1.2 * 49.63 \text{ N} = \mathbf{65.5 \text{ N}}$$

$$P = X * F_{r \text{ dynamic}} + Y * F_{a \text{ dynamic}} = 0.56 * 22.98 + 1.55 * 65.5 = \mathbf{114.39 \text{ N}}$$

Bearing Life

$$L_{10} = \left(\frac{C}{P} \right)^{\frac{10}{3}} = \left(\frac{27000}{114.3 \text{ N}} \right)^{\frac{10}{3}} = 81267895.58 \text{ revolutions}$$

In hours it becomes;

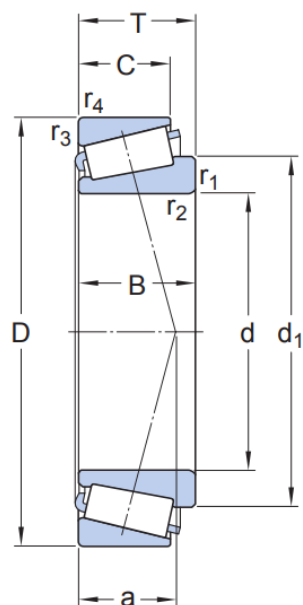
$$650 \text{ rpm} * 60 \text{ min} = 39000 \text{ rev/hour}$$

$$L_{10h} = \frac{L_{10} * 10^6}{39000} = \frac{81267895.58 * 10^6}{39000} = \mathbf{2083792194 \text{ hours}}$$

Driving Shaft Bearing (Bearing A)

Dimension series

2FB



Dimensions

d	15 mm	Bore diameter
D	42 mm	Outside diameter
T	14.25 mm	Total width
d_1	≈ 27.384 mm	Shoulder diameter of inner ring
B	13 mm	Width of inner ring
C	11 mm	Width of outer ring
$r_{1,2}$	min. 1 mm	Chamfer dimension of inner ring
$r_{3,4}$	min. 1 mm	Chamfer dimension of outer ring
a	9.45 mm	Distance side face to pressure point

Figure 8: Bearing 30302

Equivalent Load;

For bearing A, the dynamic loading is **19300N**, from the dataset, these are also the load factors given,

$$X = 0.56$$

$$Y = 1.55$$

But to account for real life scenarios, we can use other factors:

$$\text{Application Load } f_0 = 1.1$$

$$\text{Dynamic Radial } f_{rd} = 1$$

$$\text{Operating Temp } f_{tr} = 1$$

$$\text{Dynamic Axial } f_{ad} = 1.2$$

Then we can use these to calculate the dynamic load for **bearing 32204**

$$F_{r \text{ dynamic}} = f_0 * f_{rd} * f_{tr} * f_{ad} * F_r = 1.1 * 1 * 1 * 1.2 * 40.74N = \mathbf{53.77N}$$

$$F_{a \text{ dynamic}} = f_0 * f_{rd} * f_{tr} * f_{ad} * F_a = 1.1 * 1 * 1 * 1.2 * 49.63N = \mathbf{65.5 N}$$

$$P = X * F_{r \text{ dynamic}} + Y * F_{a \text{ dynamic}} = 0.56 * 53.77 + 1.55 * 65.5 = \mathbf{131.63 N}$$

Bearing Life

$$L_{10} = \left(\frac{C}{P} \right)^{\frac{10}{3}} = \left(\frac{19300}{131.63} \right)^{\frac{10}{3}} = 16619064.52 \text{ revolutions}$$

In hours it becomes;

$$650\text{rpm} * 60 \text{ min} = 39000 \text{ rev/hour}$$

$$L_{10h} = \frac{L_{10} * 10^6}{39000} = \frac{2177648.7 * 10^6}{39000} = \mathbf{426129859.5 \text{ hours}}$$

Driven Shaft Bearing D (floating)

7204B

Single row or matched pairs of angular contact ball bearings

PRODUCT DEFINITION	
Brand	NTN
d - Internal diameter	20 mm
D - External diameter	47 mm
B - Bearing/Inner ring width	14 mm
a - Charge load application point	21,5 mm
- Contact angle	40 °
rs - Min fillet radius	1 mm
r1s - Min fillet radius	0,6 mm
Radial clearance class	CN
Mass	0,102 kg

PRODUCT PERFORMANCE	
C - Dynamic load	14,7 kN
C0 - Static load	7,7 kN
Cu - Fatigue limit load	0,605 kN
N lim - Oil lubrication limit speed	20000 tr/min
N lim - Grease lubrication limit speed	15000 tr/min
Tmin - Min operating temperature	-40 °C
Tmax - Max operating temperature	120 °C

ABUTMENT	
da min - Min shoulder diameter IR	25,5 mm
db min - Min IR shoulder diameter	24,5 mm
Da max - Max shoulder diameter OR	41,5 mm
Db max - Max OR shoulder diameter	42,5 mm

Figure 9: Bearing 7204b

Equivalent Load:

For bearing D, the dynamic loading is **19500N**, from the dataset, these are also the load factors given,

$$X = 0.56$$

But to account for real life scenarios, we can use other factors:

$$\text{Application Load } f_0 = 1.1$$

$$\text{Dynamic Radial } f_{rd} = 1$$

$$\text{Operating Temp } f_{tr} = 1$$

$$\text{Dynamic Axial } f_{ad} = 1.2$$

Then we can use these to calculate the dynamic load for **bearing 32204**

$$F_{r \text{ dynamic}} = f_0 * f_{rd} * f_{tr} * f_{ad} * F_r = 1.1 * 1 * 1 * 1.2 * 91.8N = \mathbf{121.2 \text{ N}}$$

$$P = X * F_{r \text{ dynamic}} = 0.56 * 121.2N = \mathbf{67.8N}$$

Bearing Life

$$L_{10} = \left(\frac{C}{P} \right)^3 = \left(\frac{19500}{121.2N} \right)^3 = 4164817.5 \text{ revolutions}$$

In hours it becomes;

$$650\text{rpm} * 60 \text{ min} = 39000 \text{ rev/hour}$$

$$L_{10h} = \frac{L_{10} * 10^6}{39000} = \frac{4164817.5 * 10^6}{39000} = \mathbf{106790192.1 \text{ hours}}$$

Driven Shaft Bearing B

Overview

Dimensions

Bore diameter	15 mm
Outside diameter	42 mm
Width	13 mm
Contact angle	40 °

Performance

Basic dynamic load rating	13 kN
Basic static load rating	6.7 kN
Reference speed	22 000 r/min
Limiting speed	20 000 r/min

Properties

Contact type	Normal contact (two-point contact)
Number of rows	1
Locating feature, bearing outer ring	None
Ring type	One-piece inner and outer rings
Cage	Non-metallic
Matched arrangement	No
Universal matching bearing	No
Axial internal clearance	Not applicable
Tolerance class	Normal
Material, bearing	Bearing steel
Coating	Without
Sealing	Without
Lubricant	None
Relubrication feature	Without

Logistics

Product net weight	0.081 kg
eClass code	23-05-08-03
UNSPSC code	31171531

Figure 10: Bearing 7302b

Equivalent Load;

For bearing B, the dynamic loading is **14800N**, from the dataset, these are also the load factors given,

$$X = 0.56$$

But to account for real life scenarios, we can use other factors:

$$\text{Application Load } f_0 = 1.1$$

$$\text{Dynamic Radial } f_{rd} = 1$$

$$\text{Operating Temp } f_{tr} = 1$$

$$\text{Dynamic Axial } f_{ad} = 1.2$$

Then we can use these to calculate the dynamic load for **bearing 32204**

$$F_{r \text{ dynamic}} = f_0 * f_{rd} * f_{tr} * f_{ad} * F_r = 1.1 * 1 * 1 * 1.2 * 115N = \mathbf{151.8 N}$$

$$P = X * F_{r \text{ dynamic}} = 0.56 * 151.8N = \mathbf{85 N}$$

Bearing Life

$$L_{10} = \left(\frac{C}{P}\right)^3 = \left(\frac{14800}{85N}\right)^3 = 5278716.9 \text{ revolutions}$$

In hours it becomes;

$$650\text{rpm} * 60 \text{ min} = 39000 \text{ rev/hour}$$

$$L_{10h} = \frac{L_{10} * 10^6}{39000} = \frac{5278716.9 * 10^6}{39000} = \mathbf{135351714.7 \text{ hours}}$$

3.6 Parallel Key connection

Using a parallel key connection is standard. It helps to connect the gear hub and the shaft to allow transmission of torque without slipping.

The driving shaft, the one connecting to the motor, will be 15mm in diameter. The driven shaft will be 20mm. Using DIN 6558 the recommended key length for the driving shaft is 5mm x 5mm and a depth of 3mm and the depth of the groove in the hub is 2.3 mm and for the driven shaft is 6mm x 6mm with a depth of 3.5mm and groove depth of 2.8mm.

An interference fit is formed between the key and the shaft. The table of the dimensions is shown below.

shaft diameter d from ... to	cross section of the key b * h	depth of groove in shaft t ₁	depth of groove in hub t ₂
10 ... 12	4 x 4	2,5	1,8
12 ... 17	5 x 5	3	2,3
17 ... 22	6 x 6	3,5	2,8
22 ... 30	8 x 7	4	3,3
30 ... 38	10 x 8	5	3,3
38 ... 44	12 x 8	5	3,3
44 ... 50	14 x 9	5,5	3,8
50 ... 58	16 x 10	6	4,3
58 ... 65	18 x 11	7	4,4
65 ... 75	20 x 12	7,5	4,9
75 ... 85	22 x 14	9	5,4
85 ... 95	25 x 14	9	5,4
95 ... 110	28 x 16	10	6,4
110 ... 130	32 x 18	11	7,4
130 ... 150	36 x 20	12	8,4
150 ... 170	44 x 22	13	9,4
170 ... 200	45 x 25	15	10,4

Figure 11: Dimension of keys

Firstly, we calculate the permitted contact pressure between the key and the keyway. For the shaft, we have opted to use C45 steel and for the hub (the gear) we have opted to choose 42CrMo4 Steel.

We can find the pressure for the shaft using the formula:

$$p_{per} = \frac{R_b}{S_F}$$

where

R_b is the allowable yield strength, **here R_b is $\approx 245 \text{ MPa}$**

S_F is the Safety Factor, **here $S_F = 1.5$**

$$p_{per} = \frac{R_b}{S_F} = \frac{245}{1.5} = 163.33 N/mm^2$$

And for the hub,

$$p_{per} = \frac{R_b}{S_F}$$

where

R_b is the allowable yield strength, **here R_b is $\approx 500 MPa$**

S_F is the Safety Factor, **here $S_F = 2$**

$$p_{per} = \frac{R_b}{S_F} = \frac{500}{2} = 250 N/mm^2$$

Using method C, we have to ensure that the loaded length of the key is 1.3 * the diameter of the shaft, this means our key must be at least 52mm. According to this method, the average contact pressure, P_{avg}

$$p_{av} \approx \frac{F}{A} = \frac{F_t * K_\lambda}{l_l * (h - t_1) * n * \varphi} \leq p_{per}$$

with;

p_{av} : average contact pressure in $[N/mm^2]$

F_t : tangential force for the key at the joined diameter in $[N]$

K_λ : factor to consider the pressure distribution number of keys, **here K_λ is 1**

φ : factor to consider the load distribution over the number of keys, **here $\varphi=1$**

n : number of keys, **here $n=1$**

l_l : loaded length of the key in $[mm]$

h : height of the key in $[mm]$

t_1 : depth of the groove in the shaft in $[mm]$

We can then find the tangential force for the **drive shaft**, F_{t1}

$$F_{t1} = \frac{2 * 5.9 * 10^3 Nmm}{15mm} = 786.67 N$$

Then to find loaded length for the drive shaft, l_l , we use the equation:

$$l_l = \frac{F_{t1} * K_\lambda}{p_{per} * (h - t_1) * n * \varphi} = \frac{786.67 N * 1}{163.33 Nmm^2 * (5 - 3) * 1 * 1} = 2.40mm$$

Similarly for the hub, l_h , we get

$$l_h = \frac{F_{t1} * K_\lambda}{p_{per} * (h - t_1) * n * \varphi} = \frac{786.67N * 1}{250Nmm^2 * (5 - 2.3) * 1 * 1} = 1.17mm$$

Since $1.3 * d = 1.3 * 15 = 19.5mm$ which is not bigger in this case, so it works. Thus, our length of the key will be $\approx 20mm$

And for the **driven shaft**, F_{t2}

$$F_{t2} = \frac{2 * 8.9 * 10^3 Nmm}{20mm} = 890 N$$

Then to find loaded length for the drive shaft, l_l , we use the equation:

$$l_l = \frac{F_{t1} * K_\lambda}{p_{per} * (h - t_1) * n * \varphi} = \frac{890N * 1}{163.33Nmm^2 * (6 - 3.5) * 1 * 1} = 2.18mm$$

Similarly for the hub, l_h , we get

$$l_h = \frac{F_{t1} * K_\lambda}{p_{per} * (h - t_1) * n * \varphi} = \frac{890N * 1}{250Nmm^2 * (6 - 2.8) * 1 * 1} = 1.11mm$$

And, since it is less than $1.3 * 20 = 26mm$, it works, thus method c works here.

Standard 5x5x20 keys and 6x6x30 are available, thus they are selected for this case.

3.7 Sealing and housing

Lubrication plays a critical role in the longevity and reliability of any bearing system. It reduces friction between the rolling elements, helps dissipate heat, and provides a protective barrier against contaminants. Without proper lubrication, bearings are prone to overheating, wear, and premature failure.

For this bevel gear system, **multi-purpose lithium-based grease** is selected. It is ideal for medium-speed, moderate-load applications, offering excellent thermal stability, water resistance, and strong adhesion to metal surfaces. This type of grease ensures consistent performance over time with minimal maintenance, making it especially suitable for enclosed gearbox systems.

Sealing is a standard and essential feature in bevel gearboxes. It ensures that the external environment does not interfere with the internal components of the gearbox, particularly the bearings and lubrication. In this design, **radial shaft seals and rubber lip seals** are integrated around the shafts. These seals prevent dirt, moisture, and debris from entering the housing while also helping retain the lubricant inside, thus enhancing the overall efficiency and durability of the system.

For the housing, **cast iron** was selected for its strength, vibration damping properties and its ability to dissipate heat. It is also corrosion resistant and widely available. It is easy to machine and allows for complex design.

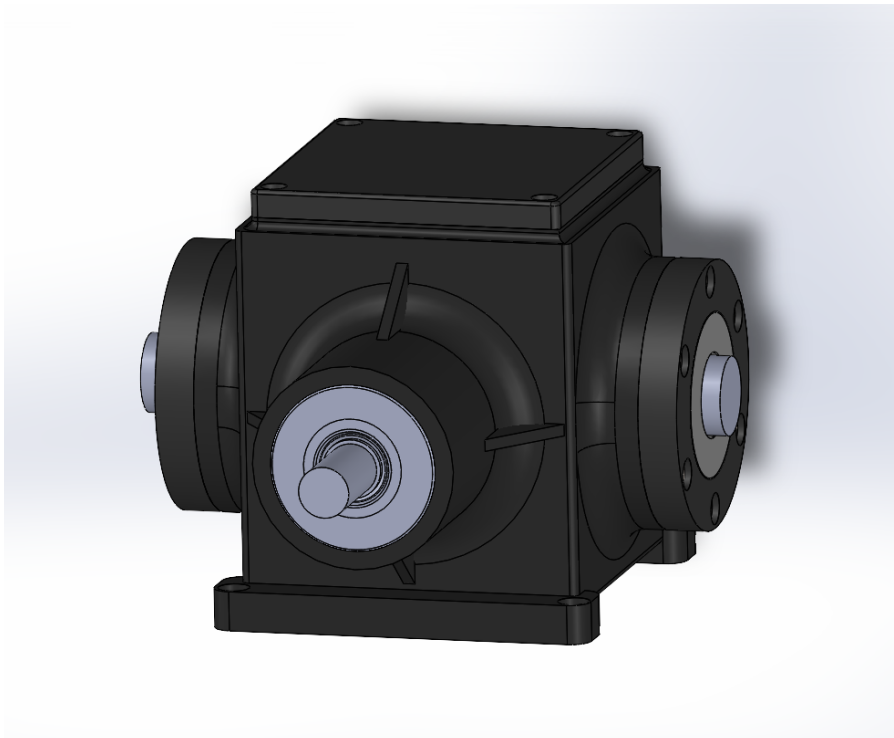


Figure 12: Bevel Gearbox

3.8 Interference check

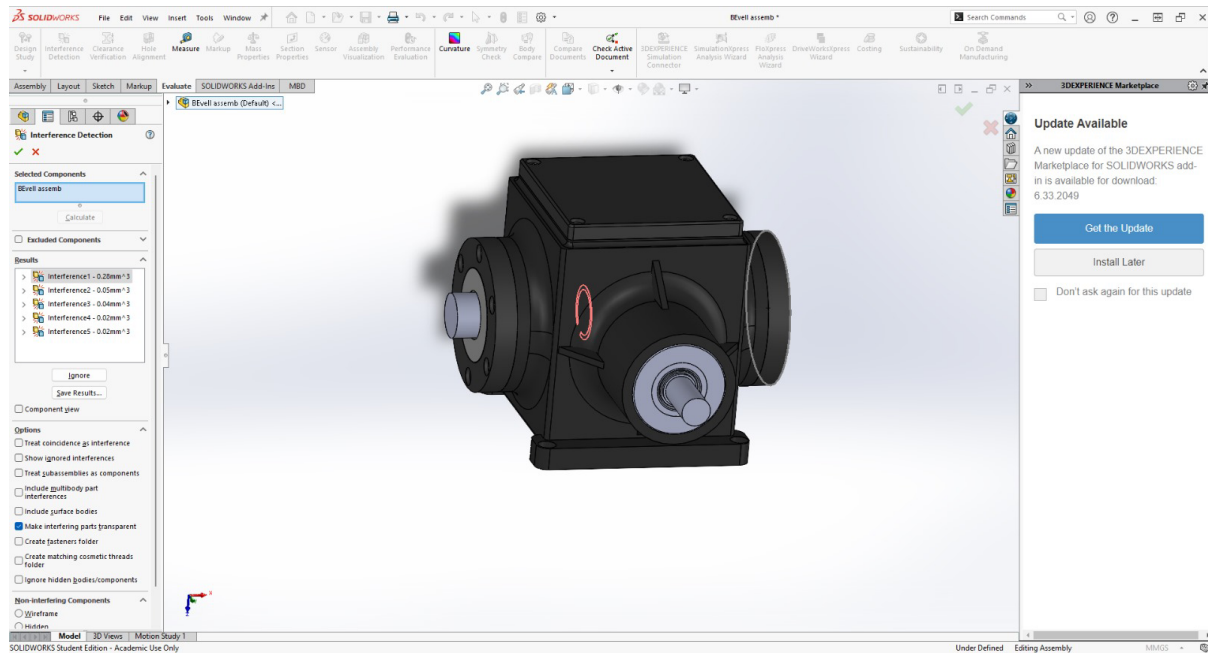
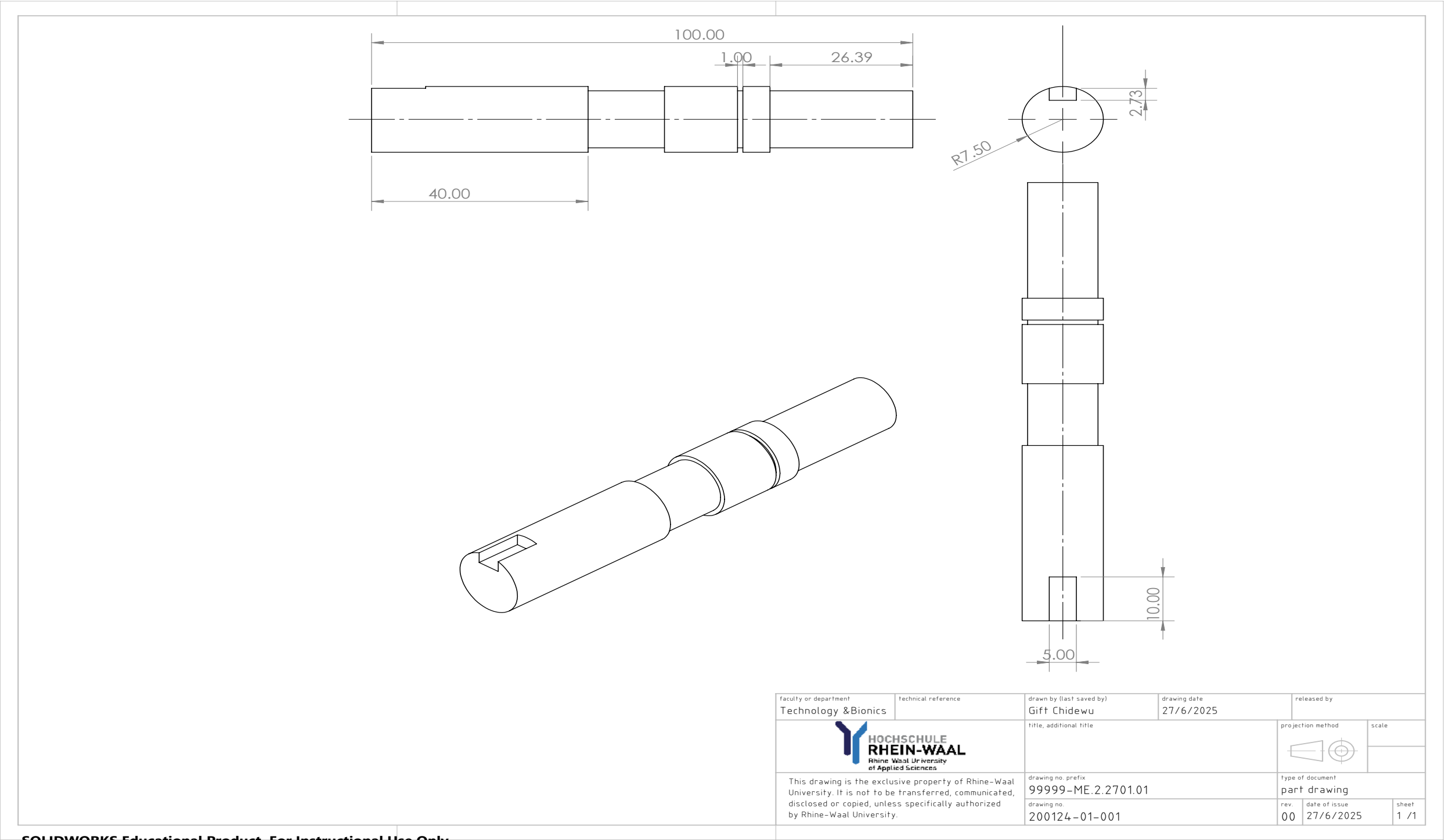


Figure 13: Interference Check

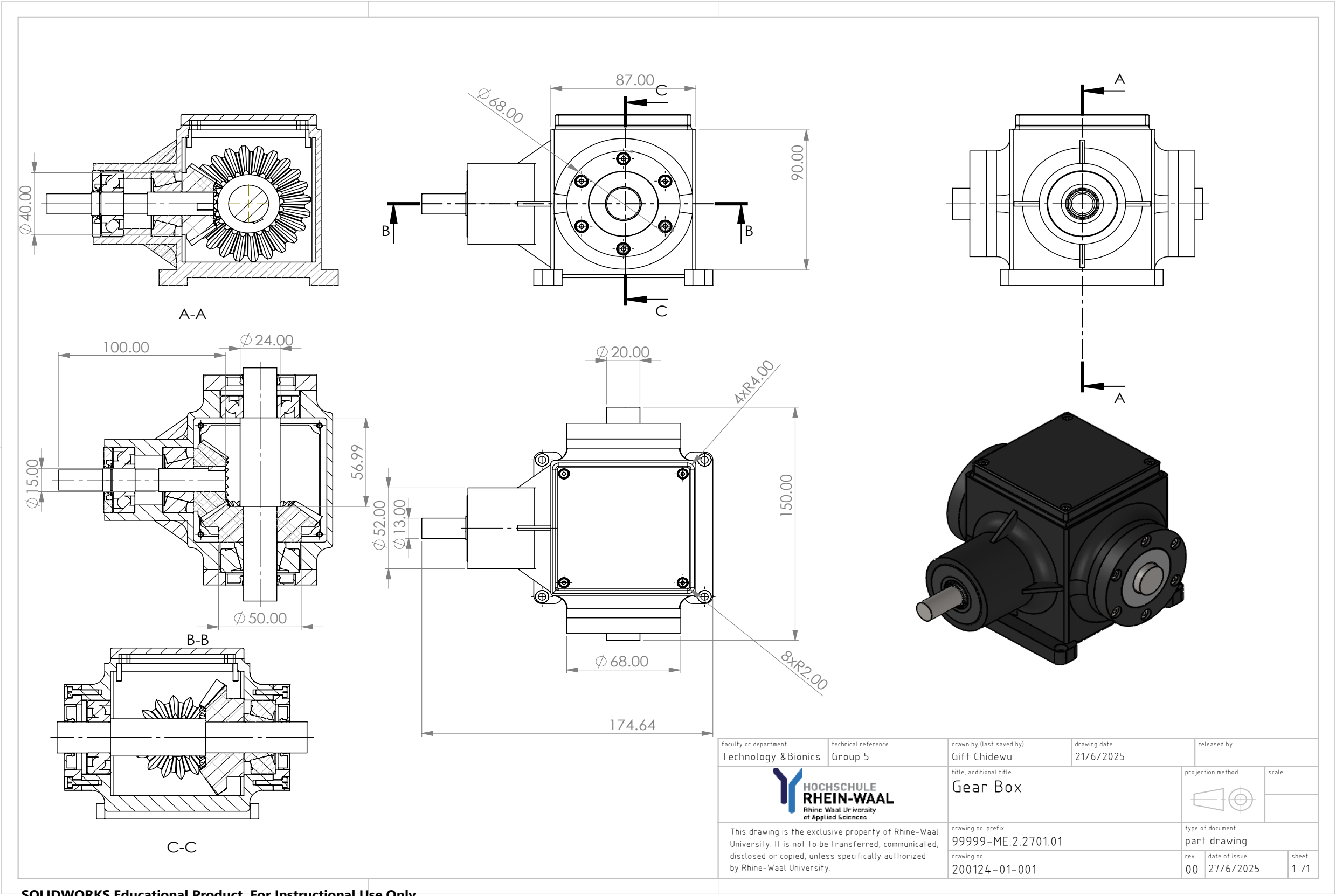
4 Drawing

4.1 Shaft drawing



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4.2 Assembly Drawing



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5 Conclusion

Bringing this project together was both a rewarding and challenging experience. Applying the theoretical knowledge from our coursework to a real-world mechanical system gave us valuable insight into the practical side of engineering.

It was especially satisfying to see how different concepts from force calculations to material selection integrated into a single, functional product. The process has definitely deepened our appreciation for mechanical design and reignited our passion for engineering.

The final gearbox design meets all the defined requirements. It is simple, cost-effective, and easy to assemble, while maintaining functional reliability. All critical calculations, from gear forces to bearing selection, were carefully carried out to ensure the design performs as intended. Selected components, such as the bearings and housing materials, were chosen to align with these results.

6 Appendix

1. https://maedlernorthamerica.com/partshop/bevel-gear-made-of-steel-module-4-16-teeth-i1-51-milled-pn-36184800/?srsltid=AfmBOor-b8YWzu7O_oMI7XLmADH-kA1rK0n4yUY-buyim0iM65lCd4yrh
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